

Seeing Human Rights at Sea: How to Align Tech Development with the Needs of Maritime Human Rights Investigators and Affected Communities

Jamie Hancock*
The Alan Turing Institute
London, United Kingdom
jhancock@turing.ac.uk

Anjali Mazumder
The Alan Turing Institute
London, United Kingdom
jatinder.singh@cl.cam.ac.uk

Ruoyun Hui
The Alan Turing Institute
London, United Kingdom
rhui@turing.ac.uk

Jatinder Singh
University of Cambridge
Cambridge, United Kingdom
The Alan Turing Institute
London, United Kingdom
amazumder@turing.ac.uk

ABSTRACT

Significant effort has been devoted to developing tech tools that leverage data to assist human rights investigations at sea. It is unclear, however, how users incorporate data and data-related tools into their investigations, nor what is needed for tech developments to contribute more value. We present the preliminary findings of a study which used workshops and interviews to examine the practices, experiences, and views of experts who perform maritime human rights investigations. Following a thematic analysis, we find that while there is scope for developers and designers to become involved, technologists' enthusiasm for cutting-edge solutions appears detached from the context and experience of affected communities. The result is a saturated but opaque market for under-used tech solutions, while communities' low-tech needs remain unmet. We suggest developers can best contribute to the fight against human rights abuses at sea by adopting deeper forms of engagement that promote communities' agency.

CCS CONCEPTS

• **Information systems** → *Open source software*; **Geographic information systems**; **Online analytical processing**.

KEYWORDS

human rights, human rights technology, maritime, labour, exploitation, migration, OSINT, data management, geolocation, remote sensing

ACM Reference Format:

Jamie Hancock, Ruoyun Hui, Anjali Mazumder, and Jatinder Singh. 2024. Seeing Human Rights at Sea: How to Align Tech Development with the Needs of Maritime Human Rights Investigators and Affected Communities.

*Corresponding author: jhamcock@turing.ac.uk

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3651081>

In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 13 pages. <https://doi.org/10.1145/3613905.3651081>

1 INTRODUCTION

Data and data-driven tools are gaining traction within investigations into human rights abuses at sea. From tracking human trafficking using vessels' geolocation data [52, 57, 70, 90] to documenting the mistreatment of migrants and asylum seekers in small boats using social media data [31, 45, 115], practitioners are turning to a plethora of data sources and analytical tools. Excitement for using data and tech has also been accompanied by concerns about invasions of privacy, loss of contextual awareness, and the reproduction of global power imbalances, amongst other issues [6, 45, 46, 53, 65, 76, 89, 109, 119]. However, most discussions in the literature are about cutting edge proof-of-concepts, theoretical concerns, or the general usage of remote sensing data. Little empirical research exists on the experiences of users of data for maritime human rights investigations, or how their perspectives can guide future tech development.

This paper addresses that gap by exploring the question: *how are data and data-related tools used in this context – and where might they fall short?* Here, we adopt a broad definition of data to include unstructured and analogue sources of information, such as witness testimony. We examine users' engagement with any tech tools used to collect, process and interpret data. Our results are derived from a desk-based review and a thematic analysis of transcripts from workshops and interviews with domain experts (N = 10), conducted in 2023. We detail the diversity of actors conducting maritime human rights investigations and their needs when it came to data and tools. We find that users' needs were shaped by a common feature of maritime environments: data tends to be scarcer and of lower quality than on land. Critically, we highlight users' concerns over how tech developers lacking expertise in maritime human rights could enter the space and crowd the market with cutting-edge technologies, whilst the 'low-tech' needs of communities affected by rights abuses went unmet. Rather than prioritising large-scale, top-down surveillance or more complex tech solutions in an already-saturated market, our findings suggest tech developers can best contribute by

adopting deep, localised forms of engagement that support users and other communities to exercise their agency.

The paper is laid out as follows. Section 2 briefly reviews the context around human rights abuses at sea, and the shifts brought by data and tech; Section 3 describes our study design and analyses; Section 4 reports findings from our workshops and interviews on the data-related practices as well as the needs and concerns our participants articulated; we conclude with a discussion on how the findings inform future development effort in Section 5.

2 BACKGROUND

We now provide context on what human rights issues arise at sea, the roles data and tech have come to play within said investigations, and the criticisms and ethical concerns that have been raised about these developments.

2.1 Data and tech have become key to human rights practice

Data and tech have grown in prominence within human rights work during recent years. According to McPherson [79], the shift has “creat[ed] a wealth of new opportunities for the field” and has given technologists a significant role to play (see also [80]). Prior studies have discussed at least six ways data and data-related tools are being used within human rights practice. These are: (1) gathering and analysing data during research, investigations, and fact-finding [22, 55, 59–61]; (2) developing aggregate statistics to observe trends and construct ‘indicators’ of the presence of rights violations or related conditions [3, 35, 63, 81, 113]; (3) presenting data as evidence during court cases [34, 47, 117]; (4) communicating about human rights issues through data visualisation and other techniques [9, 26, 92]; (5) performing administrative activities, such as data archival and case management [7, 19, 24, 72]; (6) developing policy and legal proposals [1]. Because our study focuses on how human rights practitioners identify and intervene in incidents at sea, this paper primarily discusses on the first two of these areas – data within investigations alongside data for trend and risk analysis.

2.2 Human rights abuses at sea are challenging to investigate

Growing attention has been paid to the systematic abuses of human rights at sea, culminating with the Geneva Declaration on Human Rights at Sea (GDHRS) (Mar 2022) [50]. The majority of people at sea are fishers. Labour exploitation on fishing vessels – involving deception, wage deduction, forced confinement, excessive working hours, physical and psychological abuses, and even murder – has received significant scrutiny by research and media (e.g. [27, 41, 49, 71, 106]). Offending vessels frequently also engage in illegal, unregulated, or undetected (IUU) fishing [104], threatening marine ecosystems and coastal communities’ livelihoods. Furthermore, these same sea routes can be taken by refugees and human traffickers. Like other areas in human rights, human rights work at sea shares overlaps with other domains of research and practice, such as maritime security, business, criminal justice, environmental conservation, and humanitarian action.

Maritime human rights work is challenging due to the remoteness and hard-to-observe nature of events that transpire on the

oceans. A lack of physical access, data availability, and/or necessary permissions across jurisdictions can make investigations difficult to pursue. As the GDHRS states, there is “a perception of ‘sea blindness’ reflecting the often-cited misconception that what happens at sea is out of sight and therefore out of mind” [50].

2.3 Data is used increasingly to identify hidden activities at sea

Better access to data and tools for analysing data have been proposed as solutions to some of the challenges human rights practitioners face, especially regarding a lack of direct knowledge of events at sea. There is a burgeoning literature on how data and technologies might be used to identify maritime incidents. The majority of attention appears to have gone to uses of remote sensing data to reveal the location of vessels across time. These data types include (but are not limited to): (1) Automatic Identification System (AIS) data transmitted by vessels [14, 16, 23, 30, 64, 66, 77]; (2) radar and Synthetic-aperture radar (SAR) data [12, 90]; (3) imagery captured by satellites [23, 73, 94, 108] or unmanned aerial vehicles [58]. Meanwhile, user generated data (e.g. social media data) [70] and openly available information online (i.e., Open Source Intelligence; OSINT from now on) has been proposed as an alternative pathway to identifying cases [97]. Such data can provide closer access to affected individuals and communities via the internet. Other studies have made use of state-level indices on fishery, trade and governance [29, 67, 88]; vessel registry data [87]; Port State Control inspection records [88, 120]; past offences recorded on illegal, unreported, and unregulated (IUU) fishing lists [67]; and the distribution and catch record of marine species [33, 74, 102]. These statistics are fed into a variety of applications on offer, including automated anomaly detection [66, 96], assessments of vessel and worker behaviours [56], and risk scoring [30, 62, 85]. Most of the studies to date, however, represent proofs-of-concept. The current literature lacks sufficient information about if or how the data and technologies discussed above have been implemented beyond research environments. As a consequence, it is hard to judge whether such research and development has had practical benefits.

2.4 Concerns exist about uses of data and tools

Criticisms have been levied over the use of data sources, methods, and technologies within human rights practice – including maritime rights investigations. The military origin and embedded surveillance gaze of satellite imagery have led to concerns about becoming detached from local context and local needs, enabling coercive interventions, and perpetuating a neo-colonial, technocratic gaze upon the Global South [6, 46, 65]. Technologies developed in the civic space might in turn become repurposed for use by the states or business actors [119]. Regarding Open Source Investigations (OSI) and OSINT, Rahman and Ivens have noted the dangers inherent when rights investigators adopt “the same [digital surveillance] methods that can be used by malicious actors” [91]. These methods can enable investigators to expose bystanders’ personal information in reporting, even if unwittingly, and put survivors of abuses at risk of additional harms by revealing their identities [21]. Furthermore, biases, and human misinterpretations may emerge throughout the analytics pipeline [75–77, 109]. Hence, technologists

need to be mindful of both the potential benefits tech may bring *and* the legitimate concerns held by professionals. Yet without more evidence on how practitioners conduct their data-related activities and interact with technologies, it is difficult to assess whether these concerns remain theoretical within a maritime context.

2.5 Related work

There is a growing subset of HCI literature related to human rights protection. Most studies fall under one of two main themes: 1) designing with considerations for marginalised or vulnerable communities [15, 101, 107]; 2) enhancing telecommunication security for activists, journalists, and sources [17, 20, 78]. However, the effectiveness and implications of the latest tech developments to aid human rights investigations has been rarely examined (though see [18] for a discussion on ethical risks in AI-powered tools to address human trafficking identified through a systematic literature review). With this paper, we hope to ground our reflections of data and data-related tool design in the experience of experts conducting maritime human rights investigations. Our results are intended to help developers and technologists address misalignments and tensions in the development ecosystem for investigating incidents at sea. By documenting practitioners' perspectives in such a crucial moment – when tech advancements promise to illuminate a vast, opaque space which spans diverse entities – our findings will also be of value to designers and developers seeking to leverage emerging data and tech to support investigations on land and online.

3 METHODOLOGY

We conducted workshops and interviews with practitioners (N = 10) in summer 2023. To draw out common themes, we applied thematic analysis to the transcripts using Nvivo, a qualitative analysis tool.

3.1 Practitioner engagements

Our study sought to develop an account of perspectives from experts with a diverse range of backgrounds and skill sets. We therefore sought to sample for a range of participants using two non-probability sampling strategies: purposive sampling and snowball sampling [93, 103]. Purposive sampling involves a researcher actively seeking out participants based on selection criteria [93, 103] – which, in our case, meant involvement in investigating or reporting on human rights abuses at sea using some kind of data. This enabled us to recruit participants who were particularly knowledgeable and/or prominent in the field. Snowball sampling, meanwhile, is a method wherein participants will refer researchers on to other potential participants. Snowball sampling is useful for leveraging existing participants' social networks to access to new sources and build trust [93, 103].

We drew on our literature review to devise our selection criteria, which required participants to have had relevant domain expertise (e.g., automated vessel identification) and had practical experience contributing to maritime human rights investigation and documentation projects. Our team sought participants who had been involved in projects or organisations which had received significant public attention. We looked for experts with backgrounds from a range of relevant sectors (e.g., journalism; NGOs), regardless of the type of human rights violation they examined, and reached out

to both individuals and organisations. Our aim was to give equal representation to the different sectors, skill sets, and areas of focus across the sector; as part of this, we reached out to multiple candidates from all sectors we identified. In total, we tried to contact 36 individuals and 11 organisations. This resulted in 10 individual participants. Table 1 summarises each participant's background and mode of engagement.

We ran two formats for data collection. First, we conducted two workshops (total N = 8) lasting around 90 minutes each. Then, due to scheduling issues, we chose to run two semi-structured interviews with individuals who would have otherwise not been able to participate (N = 2). These each took just under an hour. We discuss the limitations this mix of formats raises below (3.3). Ethics approval came from the Alan Turing Institute and all participants gave informed consent. All participants were asked to follow Chatham House Rules [13]; all transcripts were anonymised prior to analysis.

3.2 Thematic analysis

We utilised thematic analysis to annotate our transcripts and identify common patterns in how participants responded to prompts. Thematic coding is based on annotating and categorising qualitative data according to thematic “codes” which “symbolically assign summative, salient, essence-capturing, and/or evocative attribute[s]” [98] (also [111, 118]). Our analysis followed an open coding approach: we devised our codes inductively rather than according to a pre-defined coding schema [95, 98]. We proceeded through four stages. (1) A researcher performed an initial coding analysis using NVivo; this was fine-grained and produced over 1000 individual codes. (2) The same researcher organised these codes into a hierarchy of categories: higher-level codes represented general categories (e.g., ‘data and information’), containing more specific codes (e.g., ‘data analysis’), some of which contained even more granular codes (e.g., ‘data analysis barriers’) [37]. This produced a set of 19 top-level codes that contained a total of 1123 lower-level codes (Table 2). (3) Another researcher independently reviewed these coding results to check for reliability; a small number of codes were added or revised based on this feedback. (4) The research team used NVivo to check for patterns in which participants were associated with which codes based on different characteristics, such as their professional background.

3.3 Limitations and mitigations

We acknowledge that our sampling strategy was not conducive to producing a statistically representative picture of the field; rather, it was meant to capture the views of individuals who were influential and/or held specialised knowledge. The process of selecting and contacting participants will inevitably have introduced bias in terms of whose perspectives are featured in our results. The heavy media focus on labour issues at sea, for example, is likely to have affected our search for candidates. We were also limited in our ability to reach potential candidates for whom English was not a primary language (though we did attempt to contact such candidates via email and onward referrals from other participants). We sought to control for researcher selection bias by seeking out as wide a range of expertise as possible. Given the variety and the number of invites we sent, we suggest the greatest source of sampling bias came from

Codename	Role	Organisation type	Attended
Participant 1	Researcher/investigator	Research NGO	Workshop
Participant 2	Researcher/investigator and data scientist	Investigative NGO	Workshop
Participant 3	Researcher/investigator	Research organisation	Interview
Participant 4	Campaigner, advocate, and investigator/researcher	Trade union	Workshop
Participant 5	Campaigner and advocate	Campaigning NGO	Workshop
Participant 6	Technologist and researcher	Tool provider	Workshop
Participant 7	Technologist	Tool provider	Workshop
Participant 8	Researcher and consultant	Consultancy	Workshop
Participant 9	Investigative journalist	Freelance	Interview
Participant 10	Researcher/investigator and data scientist	Research organisation	Workshop

Table 1: Codenames and details for study participants

participants: of the 47 individuals and organisations contacted, 23 did not reply.

We also note that different dynamics may arise in focus group discussions versus interviews, which may affect results. While focus groups allow for emergent interactions, they can feature an imbalance of participation [54, 83]. By contrast, semi-structured interviews allow for a more controlled environment where participants can speak at length; however, they preclude the interactive aspects of workshops [10, 83]. We implemented three strategies to mitigate these differences. First, we followed the same semi-structured framework of prompts for both interviews and workshops (provided in Appendix A); this was used as a presentation in workshops and an interview guide in interviews. Second, as workshop moderators, we sought to ensure each workshop attendee had the opportunity to contribute as much as they wished. Finally, in our analysis, we have endeavoured not to privilege interviewees – who were able to contribute at greater length – over workshop attendees giving equal attention to both types of participant.

These limitations leave space for future research which provides a more statistically representative view of the field (e.g. surveys using randomised or stratified sampling). Similarly, there remains a gap for work which documents the standpoints of those our sampling may have missed, such as people directly affected by rights violations at sea, non-English-speaking practitioners, and practitioners whose work has received less public attention.

4 FINDINGS RELATING TO DATA AND TOOLS

We now turn to our findings. Our discussion centres on themes relating to ‘data and information’ and ‘tools and technologies’. This is because: (1) these subjects are the most relevant for guiding future tool development and will be of particular interest to the HCI community; (2) they were the explicit focus of our research question; (3) they comprised two of the most frequently occurring categories of themes. Most other prominent thematic areas – such as ‘problems and challenges’ – had overlaps with data and technology themes, and are also touched upon below. We will return to themes are not discussed in detail (e.g., ‘law and regulation’) in future writing. Table 2 offers a breakdown of all top-level themes.

4.1 Users, data types, and practices are diverse

The maritime human rights field appeared to be a complex space, interconnecting actors of many types: affected individuals and communities; community and worker representatives; NGOs and non-profits; businesses; and state authorities (see Appendix B: Table 3). These types differed by the focus of their operations, geographic location(s), scale, funding models, access to resources, legal authority, and more. Their data-related work was said to fulfil a diverse array of aims depending on their personal, organisational, and profession’s overall objectives. Rights practitioners could: attempt to tackle abuses directly through immediate interventions; assist law enforcement; support survivors and affected communities; seek to shape policy; work to reform industry; and raise public awareness.

Within our participants’ work, four violations emerged as a core focus: (1) labour exploitation; (2) human trafficking; (3) harms against migrants and refugees, such as migrant pushbacks; and (4) state violence enacted at sea. Different abuse-types were associated with specific kinds of marine activity. Participants framed labour exploitation and modern slavery, for instance, as a problem within fishing industries (see also [68, 112]). Moreover, we heard of a wide variety of kinds of data being used. It included remote sensing data (e.g. AIS, VMS, satellite imagery), state-level statistics and indicators (e.g. economic data), and OSI (e.g. user-generated content, messaging data).

4.2 Four types of investigation

We suggest categorising maritime investigations based on patterns in the data-related methods and tools they used. Based on our participants’ accounts, our analysis grouped these into four types: (1) ‘traditional’ manual investigations; (2) digital open-source investigations (i.e. OSI/OSINT); (3) large-scale, automated case detection; (4) trend analysis and synthesis.

4.2.1 ‘Traditional’ manual investigations. These were case-based, primarily offline, built using trusted sources and witness testimony, and involved careful verification work. This was termed a “bottom-up” approach by Participant 9, an investigative journalist. The approach appeared to be largely manual and could involve using analogue data in the form of paper-based records and handwritten notes. Similarly, tools and technologies mentioned tended to

Top-level theme	No. lower-level themes	Examples of lower-level themes	Coding frequency
Data and information	182	Data access, gathering, and visibility	352
Events and issues on the ground	139	Background issues (e.g. economic factors); Treatment of workers	252
Practices, methods and approaches	154	Work process; Roles and goals; Collaborations; Managing privacy, security, and risk	251
Problems and challenges	186	Gaps and lacks; Issues with tools and technology; Problems regarding data; Bias	245
Locations	46	Countries; Regions; Oceans	131
Actors	56	Named individuals; Named organisations	130
Tools and technologies	61	Current state of tools for field; Types of tools; Uses for tools	122
Indicators and signs	53	Indicator sets; Kinds of indicator	105
Engaging with survivors and communities	32	Reasons to engage; issues when engaging; building trust	71
Outcomes	43	Improving services and support; working with law enforcement	70
Abuse types	16	Modern slavery; trafficking; labour exploitation; child abuse; migrant pushbacks	62
Knowledge and perspectives	30	Standpoints; lived experience; expertise; standards of knowledge	47
Ethics and norms	24	Ethical challenges; tensions between law and ethics; minimising harm	31
Risk	13	Risk assessment; forms of risk; risk of tool misuse	28
Needs, aspirations, and future developments	28	Improving accuracy; need for standardisation; need for map of existing tools	27
Organisation, collaboration, and coordination	8	Collaborations and partnerships; bottom up responses	22
State of the field	22	Standards; editorial control; unwritten norms and standards; trust	20
Law and regulation	15	Compliance; auditing; international law; courts; legal authority to investigate	18
Definitions	8	Defining 'data'; defining 'labour abuse'; defining 'security'	13

Table 2: Top-level themes identified by thematic analysis

be comparatively ‘low-tech’. They often related to data collection, storage, and sharing: mobile phones for contacting sources; WiFi for witnesses to create and share evidence; and satellite phones for communication. In one case, Participant 3 told us, a fishing worker had “sent literally a message in a bottle” to seek help. A short list of tools were mentioned for analysing data, such as qualitative analysis software (e.g. Max QDA). When advocating for changes in data and tools, these practitioners focused primarily on two issues: a need for better communication infrastructure to enable data sharing, and more access to existing low-tech solutions on behalf of affected communities (e.g., mobile phones).

4.2.2 Digital open-source investigations. These were manual, internet-based investigations using digital open-source data. They could thus be considered a variant of OSINT (e.g., [4]). Such work had been popularised within human rights work by NGOs like Amnesty International [2], Human Rights Watch [51], and Forensic Architecture [32]. The tools and techniques used in this field were

developed primarily for investigations on land, but their usage in the maritime contexts was said to be growing. They included standard data archival software, satellite imagery platforms, and search engines. As with more traditional investigations, corroboration and triangulation were said to be key: rather than relying on one item, it was vital to connect many pieces of data to test how much each supported the others. In one case, Participant 2 described researching the mistreatment of migrants on vessels using social media posts and photographs taken from the shore. They combined these images with satellite imagery, AIS data, and other signals data from vessels. However, some standard digital investigative methods were difficult to apply: geolocation using visual features in an image, for example, was usually unfeasible for visual evidence taken at high seas without land in sight. Similarly, the viability of satellite imagery often came down to “luck” (Participant 2). Practitioners also consulted specialised resources including databases of vessel images and registration details, such as ShipSpotting [105]

and MarineTraffic [69], alongside AIS trackers like Global Fishing Watch [39].

4.2.3 Large-scale, automated case detection. This type of investigation applied computational data analysis to large datasets from remotely collected sources. If traditional investigations were ‘bottom-up’, these were ‘top-down’. They aggregated data from remote sensing, satellite imagery, and signals detection into large databases, which they would monitor continuously. Two approaches could be taken within this: (i) a system could generate automated flags for vessels displaying unusual or suspicious behaviour, which could be passed on to relevant authorities; or (ii) the organisation could select data on a specific vessel based on a tipoff. This information could then be used by ‘filtering’ to home in on potential cases. Organisations matching this pattern included Global Fishing Watch, which was co-founded by Google [38].

4.2.4 Trend analysis and synthesis. The last type centred on what we term ‘synthesising’: gathering material produced by the other kinds of investigations and collating the results to distil further insights. Synthesisers drew on both data published about individual cases and aggregated data that captured trends. Some synthesised these high-level insights into reports for public consumption (e.g. [27, 28, 41, 42]); others would use them to calibrate statistically driven decision-making and risk-assessment tools. They would occasionally feed into more direct investigations – for example, to help identify hotspots for future investigations or to determine if known risk factors were present in a particular case.

The multiplicity we observed in investigative practices means it may not make sense to view ‘human rights investigations at sea’ as a singular kind of practice performed by a cohesive community. Instead, researchers and technologists interested in the field should be aware of the many communities involved and the many kinds of investigation they conduct. Practitioners must remain cognisant that all communities come with their own nuances and needs.

4.3 Problems with data: scarcity, access, and quality

Although all four types of investigation made use of data to different extents, our interlocutors raised numerous problems regarding data – ranging from data not being collected and low data quality to difficulties they faced when trying to access or share the data. Most of all, maritime human rights work was “a data poor environment”, in Participant 1’s words. This was “despite everything that [had] happened over the last decade” in terms of technical developments. Participant 2 concurred: “data sparseness [was] always an issue” at sea. The extremely low transparency of the seafood supply chain was also mentioned by several participants.

Even when data existed, inaccessibility, incompleteness, and/or a lack of quality were said to severely restrict its value. This could feed into biases if the analysts are not aware. For example, there was a danger for investigators who depended on AIS to “miss non-AIS enabled vessels” (Participant 10). “This [was] particularly an issue in developing countries [...] on small vessels that we can’t capture using our current tools”, as AIS is not mandatory in all jurisdictions for all vessel types and is prone to manipulation and spoofing (see also [11, 86, 110, 116]). Participant 1 said that incomplete data led

to “default high risk for practically everything” when they initially tried to incorporate quantitative data sources. They had to remodel by “doing assessments that are entirely descriptive” instead to make the tool informative. These comments align with findings from studies suggesting that “in recent years [...] the largest cases of illegal fishing and forced labour were by fleets that mostly did not use AIS devices” [86]. Overarching organisational or political goals were said to sometimes prevent data access. For example, VMS data were generally gatekept by state actors. Participant 3 mentioned an occasion when “the [Peruvian] government actively denied us [VMS] data because they didn’t want us to [...] link any kind of satellite data to labour exploitation.” Participant 1 also remarked on some governments’ reluctance to acknowledge problems: “it’s an embarrassment [for them].”

These data issues appeared to not just limit the development of tools or the ability for practitioners to identify cases. If developers lack sufficient understanding about data generation processes and potential issues within the data, there is a risk such data could undermine technological interventions’ accuracy and make investigations less effective.

4.4 People remain indispensable

Within the oceans’ “data poor environment”, according to our participants, the most useful information for investigation often did not come from top-down monitoring. Rather, the best sources were said to often be affected individuals and frontline organisations who might not have sufficient support. Yet acquiring data from human sources was a key problem. Gaining physical access was “very challenging” for Participant 3: “how do you interview workers aboard distant water fishing vessels?” Potential sources could face coercive situations preventing them from providing data – e.g. if their communication devices were taken by a captain. Additionally, people with lived experiences could be unwilling to provide testimony due to past experiences. “There are plenty of people who don’t want to share their experience [because] it’s too traumatising” (Participant 9).

Most participants emphasised the importance of empowering affected communities to document abuses. But community- or victim-led monitoring also faced obstacles: frontline organisations often did not have the appropriate tools or expertise to record and share information in a standardised way. Given the key role our participants assigned to affected communities as advocates and sources, we suggest more resources and attention need to be paid to participatory design and community empowerment [99, 100].

4.5 Technical tools: not fit for users’ purposes or lacking contextual specificity

The field of marine intelligence appeared to have experienced intense growth, with many new players entering and attempting to offer their own products. This appeared to have created tensions. In particular, we heard of concerns about developers from outside the field that lacked contextual knowledge trying to offer tools that did not solve the real problems people faced (Table 6, Appendix A). Many participants mentioned instances where technological solutions were not fit for their intended purpose within maritime human rights work. Participant 3 provided the example of cameras

on vessels. These would not be useful to “detect” the “more nuanced [problems], like social issues that are happening on the vessels”. Cameras would pick up “only the vessels that are willing to install cameras on their ships [which were] obviously not going to be the ones committing the most severe infractions”. Similarly, Participant 9 expressed concern that many purported tech solutions were “divorced from people’s [lived] experience and reality”.

A gap between the reality presented by tools and lived experience was said to exacerbate another problem: biased outputs and/or the misinterpretation of results. For digital open-source investigations, Participant 2 told us that investigators’ position as “online researchers looking at data [found] on the internet” meant it was “very easy to either miss context or misinterpret”. “One of the most important things” for their team was to “hav[e] a local partner organisation in the region to co-publish with or to provide contextual support”. Not having a partner was “oftentimes a reason [not to] investigate something”. Meanwhile, Participant 3 mentioned detection tools could “create [a] false sense of bright spots, where we say: ‘the technology is not detecting anything bad and therefore it must be OK’”.

4.6 A saturated and opaque tech market

Furthermore, the number of overlapping projects and similar tools being offered was itself said to be problematic. Consider the following statement by Participant 7, a senior figure in a company offering vessel tracking tools: “I would posit as a tool provider that more tools are absolutely not what’s needed and it would make matters worse”. The level of crowding in the for-profit and not-for-profit markets meant there were “lots and lots of other organisations doing very similar things to us and to other [tool providers]”. There were “hundreds if not thousands of organisations” now involved in offering tools and services. Participant 4 agreed, pointing to the confusion caused by the “bewildering array of NGOs active” in technical development for maritime human rights- adjacent work.

Market saturation was said to have contributed to a general lack of communication or knowledge about what data and tools even existed. As Participant 7 put it, “what’s out there is not fully known and the full set of potential requirements [for front-line practitioners] are not circulating”. There appeared to be an overall feeling of dissatisfaction with the level of engagement between tool developers and investigative practitioners.

5 DISCUSSION

Even with its limited reach, our study revealed that a diverse range of actors contribute to marine human rights investigations. These consulted a wide variety of data sources and tools, with many different approaches for incorporating them into their workflow. Data and data-related tools appear to be integral to this rapidly changing field. From improved access to ‘low-tech’ tools (e.g. WiFi, mobile phones, and other communication equipment) for survivors and at-risk communities to better access to relevant, high-quality data and the technical means to analyse it, there is certainly scope for developers and designers to become involved and contribute value. The division of labour among human rights practitioners suggests there are likely under-explored opportunities to support activities not stereotypically associated with investigation, such as

community capacity building and report processing. This echoes Deeb-Swihart et al.’s study on AI for addressing human trafficking [18], which called for more research on the larger human trafficking ecosystem beyond aiding investigations by law enforcement. For complex social issues such as human rights violations, a holistic understanding of their development and existing interventions is crucial to identifying areas to maximise impact.

Currently, there seems to be a misalignment between needs versus development efforts. The market was said to feature a large (but unknown) amount of tools positioned as technological solutions, fuelled by a rush to leverage more data and deploy more sophisticated algorithms. Apart from those produced by a few well-known providers engaged in frequent communications with practitioners, most tools were said to not be in use. This was because practitioners either did not know about them or were doubtful of their suitability. Meanwhile, the technologies being offered did not meet the ‘low-tech’ needs of communities affected by rights abuses.

Perhaps the most urgent message that emerged for technologists was a call to avoid simplistic “techno-optimism” (Participant 3): the assumption that new technologies will offer the most effective solutions to detecting human rights abuses at sea. A closely related tendency – techno-solutionism – has long been criticised for perpetuating bias and prejudice, neglecting the interconnectedness and complexity of social phenomena, sidelining diverse perspectives, and threatening moral and civic values [44, 82, 84]. Developers should be cognisant that the current scarcity and low quality of data sources poses a basic problem for analytics solutions – especially where these introduce biases. Moreover, they must have a deeper understanding of the ethical concerns and trade-offs faced by human rights practitioners in their overall effort to minimise harm and safeguard at-risk communities. If technologists respond to existing data scarcity problems by advocating for larger-scale data collection, for instance, they must account for the fact that this may represent an expansion of surveillance and may result in further harms. Collaborations will only be effective in this space when tools offered are fit for purpose and their capabilities are communicated effectively. The implications of low data quality and excessive data collection have also been examined amid attempts at datafication in other application areas [40, 43], and should be treated with caution.

For longer-term project planning, developers and designers should recognise the importance of engaging at local community levels. Frameworks such as human-centred computing and participatory design can help to develop a close and continued partnership with affected communities, ensure the effectiveness of products, and promote community autonomy and agency [8, 18, 48]. This was a theme running through our discussion on data, tools, approaches and concerns with practitioners. Communities and affected individuals often contribute the most relevant data thanks to their physical access and ability to maintain continuous monitoring. They are best placed to interpret findings based on their own lived experience and rich contextual knowledge. Community- or victim-led interventions mitigate against the risks of constituting surveillance and perpetuating further harm, and instead become an exercise of agency and empowerment. Addressing their needs may not involve cutting-edge technological solutions, and might not tackle interconnected problems across a global scale. Rather, supporting

the long-term safety, resilience and empowerment of vulnerable individuals and communities requires designers and developers to work with existing strengths and social bonds in the community [36].

6 CONCLUSION AND PATHS FORWARD

Our results provide a much-needed evaluation of the effectiveness of data and data-related tools in human rights investigations at sea. We argue that technology designs must be informed by the wider social contexts they operate in. With regard to exposing human rights abuses and safeguarding individuals, this means: appreciating people's many vulnerabilities and needs; understanding the contexts which enable and inhibit abuses; carefully examining the limitations of emerging data and technologies; and strengthening capacities that may exist within communities. We hope this preliminary study helps to guide future development efforts towards deeper engagement with local organisations and communities.

We have outlined several avenues forward for researchers and designers. For those interested in tool design and human-computer interaction, we wish to highlight the need articulated to us for accessible tools that will empower people with lived experience of maritime rights abuses. Achieving this will require careful, contextually-informed work with communities. For researchers, there seems to be an opening for research and development within a co-design, participatory, and/or emancipatory framework [5, 25, 114]. There also appears to be a specific need for a study that extensively maps the landscape of tools available. We aim to address these research gaps in future work.

ACKNOWLEDGMENTS

We would like to thank all our participants for contributing their time and expertise to this project. We thank the Associate Chair and reviewers for their helpful suggestions on improving this manuscript. We also thank Emma Kallina, Yesim Kakalic, Sarada Mahesh, and Harry Bowles for providing feedback on previous versions of this manuscript.

This work was funded by The Alan Turing Institute.

REFERENCES

- [1] Adriana Alvarado Garcia, Matthew J. Britton, Dhairya Manish Doshi, Munmun De Choudhury, and Christopher A. Le Dantec. 2021. Data Migrations: Exploring the Use of Social Media Data as Evidence for Human Rights Advocacy. *Proceedings of the ACM on Human-Computer Interaction* 4, CSCW3 (1 2021), 1–25. <https://doi.org/10.1145/3434177>
- [2] Amnesty International. 2024. About Amnesty's Citizen Evidence Lab. <https://citizenevidence.org/about-us/>
- [3] Jana Asher, David Banks, and Fritz J. Scheuren. 2008. *Statistical methods for human rights*. Springer, New York, 1–339 pages. <https://doi.org/10.1007/978-0-387-72837-7/COVER>
- [4] Rae Baker. 2019. OSINT on the Ocean: Maritime Intelligence Gathering Techniques. <https://wondersmithrae.medium.com/osint-on-the-ocean-maritime-intelligence-gathering-techniques-2ee39e554fe1>
- [5] Liam Bannon, Jeffrey Bardzell, and Susanne Bødker. 2018. Reimagining participatory design. *Interactions* 26, 1 (12 2018), 27–32. <https://doi.org/10.1145/3292015>
- [6] Mia M Bennett, Janice K Chen, Luis F Alvarez León, and Colin J Gleason. 2022. The politics of pixels: A review and agenda for critical remote sensing. *Progress in Human Geography* 46, 3 (6 2022), 729–752. <https://doi.org/10.1177/03091325221074691>
- [7] Oriana Bernasconi. 2022. Human Rights Archives. In *The International Encyclopedia of Anthropology*. Wiley, 1–11. <https://doi.org/10.1002/9781118924396.wbiea2517>
- [8] Elizabeth Bondi, Lily Xu, Diana Acosta-Navas, and Jackson A. Killian. 2021. Envisioning Communities: A Participatory Approach Towards AI for Social Good. In *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society*. ACM, Virtual Event USA, 425–436. <https://doi.org/10.1145/3461702.3462612>
- [9] Jeremy Boy, Anshul Vikram Pandey, John Emerson, Margaret Satterthwaite, Oded Nov, and Enrico Bertini. 2017. Showing people behind data: Does anthropomorphizing visualizations elicit more empathy for human rights data? *Conference on Human Factors in Computing Systems - Proceedings* 2017-May (5 2017), 5462–5474. <https://doi.org/10.1145/3025453.3025512>
- [10] Svend Brinkmann. 2017. The Interview. In *The SAGE Handbook of Qualitative Research* (5 ed.), Norman K. Denzin and Yvonna S. Lincoln (Eds.). SAGE Publications, Thousand Oaks, CA.
- [11] C4ADS. 2019. *Above Us Only Stars: Exposing GPS Spoofing in Russia and Syria*. Technical Report. C4ADS, Washington, DC. <https://c4ads.org/reports/above-us-only-stars/>
- [12] Terry Caelli, Joyanto Mukerjee, and Evan Sparks. 2018. Analytics for awareness in maritime surveillance: from data to tactical insight. <https://doi.org/10.1177/1548512918795738> 16, 2 (8 2018), 207–215. <https://doi.org/10.1177/1548512918795738>
- [13] Chatham House. 2024. Chatham House Rule. <https://www.chathamhouse.org/about-us/chatham-house-rule>
- [14] Sudhir Kumar Chaturvedi, Chan Su Yang, Kazuo Ouchi, and Palanisamy Shanmugam. 2012. Ship Recognition by Integration of SAR and AIS. *The Journal of Navigation* 65, 2 (4 2012), 323–337. <https://doi.org/10.1017/S0373463311000749>
- [15] Christine Chen, Nicola Dell, and Franziska Roesner. 2019. Computer Security and Privacy in the Interactions Between Victim Service Providers and Human Trafficking Survivors. In *28th USENIX Security Symposium (USENIX Security 19)*. USENIX Association, Santa Clara, CA, 89–104. <https://www.usenix.org/conference/usenixsecurity19/presentation/chen>
- [16] Xinqiang Chen, Jun Ling, Yongsheng Yang, Hailin Zheng, Pengwen Xiong, Octavian Postolache, and Yong Xiong. 2020. Ship Trajectory Reconstruction from AIS Sensory Data via Data Quality Control and Prediction. *Mathematical Problems in Engineering* 2020 (2020). <https://doi.org/10.1155/2020/7191296>
- [17] Alaa Daffalla, Lucy Simko, Tadayoshi Kohno, and Alexandru G. Bardas. 2021. Defensive Technology Use by Political Activists During the Sudanese Revolution. In *2021 IEEE Symposium on Security and Privacy (SP)*. IEEE, San Francisco, CA, USA, 372–390. <https://doi.org/10.1109/SP40001.2021.00055>
- [18] Julia Deeb-Swihart, Alex Endert, and Amy Bruckman. 2022. Ethical Tensions in Applications of AI for Addressing Human Trafficking: A Human Rights Perspective. *Proceedings of the ACM on Human-Computer Interaction* 6, CSCW2 (Nov. 2022), 1–29. <https://doi.org/10.1145/3555186>
- [19] Jeff Deutch. 2020. Challenges in Codifying Events Within Large and Diverse Data Sets of Human Rights Documentation: Memory, Intent, and Bias. *International Journal of Communication* 14, 0 (9 2020), 17. <https://ijoc.org/index.php/ijoc/article/view/8520>
- [20] Philip Di Salvo. 2021. Securing Whistleblowing in the Digital Age: SecureDrop and the Changing Journalistic Practices for Source Protection. *Digital Journalism* 9, 4 (April 2021), 443–460. <https://doi.org/10.1080/21670811.2021.1889384>
- [21] Sam Dubberley. 2015. Protecting the Victim's Identity. Should we be thinking about the... | by Eyewitness Media Hub | First Draft Footnotes | Medium. <https://medium.com/1st-draft/protecting-the-victim-s-identity-3b7df432ec09>
- [22] Sam Dubberley and Gabriela Ivens. 2022. *Outlining a Human-Rights Based Approach to Digital Open Source Investigations: A guide for human rights organisations and open source researchers*. Technical Report.
- [23] Daniel C Dunn, Caroline Jablonicky, Guillermo O Crespo, Douglas J McCauley, David A Kroodsma, Kristina Boerder, Kristina M Gjerde, and Patrick N Halpin. 2018. Empowering high seas governance with satellite vessel tracking data. *Fish and Fisheries* 19, 4 (7 2018), 729–739. <https://doi.org/10.1111/faf.12285>
- [24] Abigail Durrant, David S. Kirk, and Stuart Reeves. 2014. Human values in curating a human rights media archive. *Conference on Human Factors in Computing Systems - Proceedings* (2014), 2685–2694. <https://doi.org/10.1145/2556288.2557196>
- [25] Pelle Ehn, Elisabet Nilsson, and Richard Topgaard. 2014. *Making Futures*. The MIT Press, Cambridge, MA. <https://doi.org/10.7551/MITPRESS/9874.001.0001>
- [26] John Emerson, Margaret L. Satterthwaite, and Anshul Vikram Pandey. 2018. The Challenging Power of Data Visualization for Human Rights Advocacy. *New Technologies for Human Rights Law and Practice* (4 2018), 162–187. <https://doi.org/10.1017/9781316838952.008>
- [27] Environmental Justice Foundation. 2013. *Sold to the Sea: Human trafficking in Thailand's fishing industry*. Technical Report. https://ejfoundation.org/resources/downloads/Sold_to_the_Sea_report_lo-res-v2.compressed-2.compressed.pdf
- [28] Environmental Justice Foundation. 2019. *Blood and Water: Human rights abuse in the global seafood industry*. Technical Report. <https://ejfoundation.org/resources/downloads/Blood-water-06-2019-final.pdf>
- [29] Jessica H Ford and Chris Wilcox. 2019. Shedding light on the dark side of maritime trade – A new approach for identifying countries as flags of convenience. *Marine Policy* 99 (1 2019), 298–303. <https://doi.org/10.1016/j.marpol.2018.10.026>

- [30] Jessica H Ford and Chris Wilcox. 2022. Quantifying risk assessments for monitoring control and surveillance of illegal fishing. *ICES Journal of Marine Science* 79, 4 (5 2022), 1113–1119. <https://doi.org/10.1093/icesjms/fsac027>
- [31] Forensic Architecture. 2022. Drift-backs in the Aegean Sea. <https://aegean.forensic-architecture.org/>
- [32] Forensic Architecture. 2024. Forensic Architecture. <https://forensic-architecture.org/about/agency>
- [33] Timothy H Frawley, Barbara Muhling, Heather Welch, Katherine L Seto, Shui-Kai Chang, Francisco Blaha, Quentin Hanich, Myeonghwa Jung, Elliott L Hazen, Michael G Jacox, and Stephanie Brodie. 2022. Clustering of disaggregated fisheries data reveals functional longline fleets across the Pacific. *One Earth* 5, 9 (9 2022), 1002–1018. <https://doi.org/10.1016/j.oneear.2022.08.006>
- [34] Lindsay Freeman. 2019. Prosecuting Atrocity Crimes with Open Source Evidence. In *Digital Witness: Using Open Source Information for Human Rights Investigation, Documentation, and Accountability*, Sam Dubberly, A Koenig, and D Murray (Eds.). Oxford University Press, Chapter 3, 48–67.
- [35] Anne T Gallagher. 2017. What's Wrong with the Global Slavery Index? *Anti-Trafficking Review* 8 (4 2017). <https://doi.org/10.14197/atr.20121786>
- [36] Aakash Gautam, Deborah Tatar, and Steve Harrison. 2020. Crafting, Communality, and Computing: Building on Existing Strengths To Support a Vulnerable Population. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. ACM, Honolulu HI USA, 1–14. <https://doi.org/10.1145/3313831.3376647>
- [37] Graham R. Gibbs. 2007. *Analyzing Qualitative Data*. SAGE Publications, London. <https://doi.org/10.4135/9781849208574>
- [38] Global Fishing Watch. 2024. About Us. <https://globalfishingwatch.org/about-us/>
- [39] Global Fishing Watch. 2024. Transparency for a Sustainable Ocean | Global Fishing Watch. <https://globalfishingwatch.org/>
- [40] Lesley Gourlay. 2022. Surveillance and Datafication in Higher Education: Documentation of the Human. *Postdigital Science and Education* (Oct. 2022). <https://doi.org/10.1007/s42438-022-00352-x>
- [41] Greenpeace. 2018. *Misery at sea: human suffering in Taiwan's distant water fishing fleets*. Technical Report. <https://www.greenpeace.org/aotearoa/publication/misery-at-sea/>
- [42] Greenpeace. 2020. *Choppy Waters: Forced labour and illegal fishing in Taiwan's distant water fisheries*. Technical Report. <https://www.greenpeace.org/static/planet4-southeastasia-stateless/2020/03/b87c6229-2020-choppy-waters-en.pdf>
- [43] Karin Hansson, Teresa Cerratto Pargman, and Anna Dahlgren. 2020. Datafication and cultural heritage: provocations, threats, and design opportunities. (2020). https://doi.org/10.18420/ECSCW2020_WS05 Publisher: [object Object].
- [44] Janet Haven and Danah Boyd. 2020. *Philanthropy's techno-solutionism problem*. Technical Report. Data & Society Research Institute, New York. <https://knightfoundation.org/wp-content/uploads/2020/11/KF-Kettering-Philanthropy-Techno-Solution-Problem.pdf>
- [45] Charles Heller, Lorenzo Pezzani, and Maurice Stierl. 2017. Disobedient Sensing and Border Struggles at the Maritime Frontier of Europe – spheres. *Spheres* 4 (6 2017). <https://spheres-journal.org/contribution/disobedient-sensing-and-border-struggles-at-the-maritime-frontier-of-europe/>
- [46] Andrew Herscher. 2014. Surveillant Witnessing: Satellite Imagery and the Visual Politics of Human Rights. *Public Culture* 26, 3 (2014), 469–500. <https://doi.org/10.1215/08992363-2683639>
- [47] Eliot Higgins. 2023. How Open Source Evidence was Upheld in a Human Rights Court - bellingscat. <https://www.bellingcat.com/resources/2023/03/28/how-open-source-evidence-was-upheld-in-a-human-rights-court/>
- [48] Isaac Holeman and Dianna Kane. 2020. Human-centered design for global health equity. *Information Technology for Development* 26, 3 (July 2020), 477–505. <https://doi.org/10.1080/02681102.2019.1667289>
- [49] Esther Htusan and Margie Mason. 2015. More than 2,000 enslaved fishermen rescued in 6 months. *Associated Press* (9 2015). <https://www.ap.org/explore/seafood-from-slaves/more-than-2,000-enslaved-fishermen-rescued-in-6-months.htm>
- [50] Human Rights at Sea. 2022. Geneva Declaration on Human Rights at Sea. https://www.humanrightsatsea.org/sites/default/files/media-files/2022-02/GDHRAS_Jan_2022_Final_online_version_sp%20%281%29.pdf
- [51] Human Rights Watch. 2024. Digital Investigations Lab. <https://www.hrw.org/topic/technology-and-rights/digital-investigations-lab>
- [52] ILO. 2024. Using technology to eliminate forced labour on the ocean. http://www.ilo.org/global/topics/sdg-2030/goal-8/target-8-7/accelerator-lab/projects/mpfund/WCMS_856957/lang--en/index.htm
- [53] Maria Gabrielsen Jumbert. 2018. Control or rescue at sea? Aims and limits of border surveillance technologies in the Mediterranean Sea. *Disasters* 42, 4 (10 2018), 674–696. <https://doi.org/10.1111/DISA.12286>
- [54] George Kamberelis, Greg Dimitriadis, and Alyson Welker. 2017. Focus Group Research and/in Figured Worlds. In *The SAGE Handbook of Qualitative Research* (5 ed.), Norman K. Denzin and Yvonna S. Lincoln (Eds.). SAGE Publications, Thousand Oaks, CA.
- [55] P Karthikeyan and Pao-Ann Hsiung. 2022. Labour Exploitation Investigation using Satellite based Vessel Monitoring Systems. In *2022 3rd International Conference on Computing, Analytics and Networks (ICAN)*. IEEE, Rajpura, Punjab, India, 1–5. <https://doi.org/10.1109/ICAN56228.2022.10007411>
- [56] P. Karthikeyan, Tan Khang Luong, Jui An Ku, Chih Chun Chang, Hsiu Chun Hsu, and Pao Ann Hsiung. 2022. Smart Observer for Distant Water Fishing in Taiwan. In *2022 International Conference on Computational Science and Computational Intelligence, CSCI 2022*. Institute of Electrical and Electronics Engineers Inc., 1454–1460. <https://doi.org/10.1109/CSCI58124.2022.00259>
- [57] Zuzanna Klawikowska, Tomasz Ujaziowski, Michal Grochowski, and Robert Piotrowski. 2022. How High-Tech Solutions Support the Fight Against IUU and Ghost Fishing: A Review of Innovative Approaches, Methods, and Trends. *IEEE Access* 10 (2022), 112539–112554. <https://doi.org/10.1109/ACCESS.2022.3212384>
- [58] A. M. Klimkowska and I. Lee. 2017. A preliminary study of ship detection from uav images based on color space conversion and image segmentation. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences XLII-2-W6*, 2W6 (8 2017), 189–193. <https://doi.org/10.5194/ISPRS-ARCHIVES-XLII-2-W6-189-2017>
- [59] Alexa Koenig, Emma Irving, Yvonne McDermott, and Daragh Murray. 2021. New Technologies and the Investigation of International Crimes. *Journal of International Criminal Justice* 19, 1 (9 2021), 1–7. <https://doi.org/10.1093/jicj/mqab040>
- [60] Alexa Koenig and Andrea Lampros. 2020. *The Human Rights Investigations Lab: Reflections From Our Pilot Year*. Technical Report. www.cghr.polis.cam.ac.uk
- [61] Alice Kohli. 2018. Data-Driven Human Rights Investigations. In *Digital Investigative Journalism*. Springer International Publishing, Cham, 67–77. https://doi.org/10.1007/978-3-319-97283-1_7
- [62] Birgitte Krogh-Poulsen, Sara L McDonald, and Tania Woodcock. 2022. *Seafood Social Risk Tool: Identifying risk of forced labor, human trafficking, and hazardous child labor in the seafood industry*. Technical Report. Monterey Bay Aquarium. <https://www.seafoodwatch.org/our-projects/seafood-social-risk-tool>
- [63] Todd Landman. 2020. Measuring modern slavery: Law, human rights, and new forms of data. *Human Rights Quarterly* 42, 2 (5 2020), 303–331. <https://doi.org/10.1353/hrq.2020.0019>
- [64] Philipp Last, Christian Bahlke, Martin Hering-Bertram, and Lars Linsen. 2014. Comprehensive Analysis of Automatic Identification System (AIS) Data in Regard to Vessel Movement Prediction. *The Journal of Navigation* 67, 5 (2014), 791–809. <https://doi.org/10.1017/S0373463314000253>
- [65] Karen T Litfin. 2002. Public eyes: satellite imagery, the globalization of transparency, and new networks of surveillance. *Information technologies and global politics: The changing scope of power and governance* (2002), 65–88.
- [66] Changqing Liu and Xiaoqian Chen. 2013. Inference of Single Vessel Behaviour with Incomplete Satellite-based AIS Data. *The Journal of Navigation* 66, 6 (11 2013), 813–823. <https://doi.org/10.1017/S0373463313000374>
- [67] G Macfadyen, G Hosch, N Kayser, and L Tagziria. 2019. *The IUU fishing index, 2019. Poseidon Aquatic resource management limited and the global initiative against transnational organized crime*. Global Initiative. <https://globalinitiative.net/wp-content/uploads/2019/02/...>
- [68] Mary Mackay, Britta Denise Hardesty, and Chris Wilcox. 2020. The Intersection Between Illegal Fishing, Crimes at Sea, and Social Well-Being. *Frontiers in Marine Science* 7 (10 2020), 589000. <https://doi.org/10.3389/fmars.2020.589000/BIBTEX>
- [69] MarineTraffic. 2024. MarineTraffic. <https://www.marinetraffic.com/en/ais/home/centerx:-12.0/centery:25.0/zoom:4>
- [70] Melissa Marschke, Mark Andrachuk, Peter Vandergest, and Catherine McGovern. 2020. Assessing the role of information and communication technologies in responding to 'slavery scandals'. *Maritime Studies* 19, 4 (12 2020), 419–428. <https://doi.org/10.1007/s40152-020-00201-2>
- [71] Melissa Marschke and Peter Vandergest. 2016. Slavery scandals: Unpacking labour challenges and policy responses within the off-shore fisheries sector. *Marine Policy* 68 (6 2016), 39–46. <https://doi.org/10.1016/j.marpol.2016.02.009>
- [72] Mehluli Masuku. 2012. Archives, accountability, human rights and good governance : where is the nexus? . *Mousaion* (1 2012). <https://journals.co.za/doi/abs/10.10520/EJC144282>
- [73] G. Mátyus. 2013. Near real-time automatic marine vessel detection on optical satellite images. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences XL-1-W1* (5 2013), 233–237. <https://doi.org/10.5194/ISPRSARCHIVES-XL-1-W1-233-2013>
- [74] Sam McClatchie. 2021. *Distant-water industrial fishing in high diversity regions*. Technical Report. Oceanography. <https://doi.org/10.1002/essoar.10507004.1>
- [75] Yvonne McDermott, Alexa Koenig, and Daragh Murray. 2021. Open source information's blind spot: Human and machine bias in international criminal investigations. *Journal of International Criminal Justice* 19, 1 (2021), 85–105. <https://doi.org/10.1093/jicj/mqab006>
- [76] Gavin G McDonald, Christopher Costello, Jennifer Bone, Reniel B Cabral, Valerie Farabee, Timothy Hochberg, David Kroodsmá, Tracey Mangin, Kyle C Meng, and Oliver Zahn. 2021. Reply to Swartz et al.: Challenges and opportunities for identifying forced labor using satellite-based fishing vessel monitoring. *Proceedings of the National Academy of Sciences* 118, 19 (5 2021), e2104563118.

- <https://doi.org/10.1073/pnas.2104563118>
- [77] Gavin G McDonald, Christopher Costello, Jennifer Bone, Reniel B Cabral, Valerie Farabee, Timothy Hochberg, David Kroodsmas, Tracey Mangin, Kyle C Meng, and Oliver Zahn. 2021. Satellites can reveal global extent of forced labor in the world's fishing fleet. *Proceedings of the National Academy of Sciences* 118, 3 (1 2021), e2016238117. <https://doi.org/10.1073/pnas.2016238117>
 - [78] Susan E. McGregor, Elizabeth Anne Watkins, Mahdi Nasrullah Al-Ameen, Kelly Caine, and Franziska Roesner. 2017. When the Weakest Link is Strong: Secure Collaboration in the Case of the Panama Papers. In *26th USENIX Security Symposium (USENIX Security 17)*. USENIX Association, Vancouver, BC, 505–522. <https://www.usenix.org/conference/usenixsecurity17/technical-sessions/presentation/mcgregor>
 - [79] Ella McPherson. 2015. *ICTs and Human Rights Practice: A Report Prepared for the UN Special Rapporteur on Extrajudicial, Summary, or Arbitrary Executions*. Technical Report. www.cghr.polis.cam.ac.uk
 - [80] Ella McPherson, Isabel Guenet Thornton, and Matt Mahmoudi. 2019. Open Source Investigations and the Technology-Driven Knowledge Controversy in Human Rights Fact-Finding.
 - [81] Sally Engle Merry. 2016. *The seductions of quantification: measuring human rights, gender violence, and sex trafficking*. The University of Chicago Press. 249 pages.
 - [82] Stefania Milan. 2020. Techno-solutionism and the standard human in the making of the COVID-19 pandemic. *Big Data & Society* 7, 2 (July 2020), 205395172096678. <https://doi.org/10.1177/2053951720966781>
 - [83] David L. Morgan. 1996. Focus Groups. *Annual Review of Sociology* 22, 1 (8 1996), 129–152. <https://doi.org/10.1146/annurev.soc.22.1.129>
 - [84] Evgeny Morozov. 2013. *To save everything, click here: the folly of technological solutionism* (first edition ed.). PublicAffairs.
 - [85] Katrina Nakamura, Lori Bishop, Trevor Ward, Ganapathiraju Pramod, Dominic Chakra Thomson, Patima Tungpuchayakul, and Sompong Srakaew. 2018. Seeing slavery in seafood supply chains. *Science Advances* 4, 7 (7 2018). [https://doi.org/10.1126/SCIADV.1701833/SUPPL\[\]FILE/1701833\[\]SM.PDF](https://doi.org/10.1126/SCIADV.1701833/SUPPL[]FILE/1701833[]SM.PDF)
 - [86] Fernando Paolo, David Kroodsmas, Jennifer Raynor, Tim Hochberg, Pete Davis, Jesse Cleary, Luca Marsaglia, Sara Orofino, Christian Thomas, and Patrick Halpin. 2024. Satellite mapping reveals extensive industrial activity at sea. *Nature* 2024 625:7993 625, 7993 (1 2024), 85–91. <https://doi.org/10.1038/s41586-023-06825-8>
 - [87] Jaeyoon Park, Jennifer Van Osdel, Joanna Turner, Courtney M Farthing, Nathan A Miller, Hannah L Linder, Guillermo Ortuño Crespo, Gabrielle Carmine, and David A Kroodsmas. 2023. Tracking elusive and shifting identities of the global fishing fleet. *Science Advances* 9, 3 (1 2023), eabp8200. <https://doi.org/10.1126/sciadv.abp8200>
 - [88] Gohar A Petrossian, Monique Sosnowski, Dana Miller, and Diba Rouzbahani. 2020. Flags for sale: An empirical assessment of flag of convenience desirability to foreign vessels. *Marine Policy* 116 (6 2020), 103937. <https://doi.org/10.1016/j.marpol.2020.103937>
 - [89] Lorenzo Pezzani and Charles Heller. 2019. AIS Politics: The Contested Use of Vessel Tracking at the EU's Maritime Frontier. <https://doi.org/10.1177/0162243919852672> 44, 5 (5 2019), 881–899. <https://doi.org/10.1177/0162243919852672>
 - [90] M. S. Prasad, Shivani Verma, and Yulia A. Shichkina. 2023. Dark Ship detection: SAR Images. In *2023 XXVI International Conference on Soft Computing and Measurements (SCM)*. IEEE, 341–344. <https://doi.org/10.1109/SCM58628.2023.10159062>
 - [91] Zara Rahman and Gabriela Ivens. 2019. Ethics in Open Source Investigations in: Digital Witness. In *Digital Witness: Using Open Source Information for Human Rights Investigation, Documentation, and Accountability*, Sam Dubberley, Alexa Koenig, and Daragh Murray (Eds.). Oxford University Press, Oxford, Chapter 11. <https://opil.ouplaw.com/display/10.1093/law/9780198836063.001.0001/law-9780198836063-chapter-12>
 - [92] Katharina Rall, Margaret L. Satterthwaite, Anshul Vikram Pandey, John Emerson, Jeremy Boy, Oded Nov, and Enrico Bertini. 2016. Data Visualization for Human Rights Advocacy. *Journal of Human Rights Practice* 8, 2 (7 2016), 171–197. <https://doi.org/10.1093/JHUMAN/HUW011>
 - [93] Tim Rapley. 2014. Sampling Strategies in Qualitative Research. In *The SAGE Handbook of Qualitative Data Analysis*, Uwe Flick (Ed.). SAGE Publications, London, Chapter 4. <https://doi.org/10.4135/9781446282243>
 - [94] T. Reize, R. Müller, and R. Kiefl. 2013. An Interactive Graphical User Interface for Maritime Security Services. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences* XL-7-W2, 7W2 (10 2013), 201–206. <https://doi.org/10.5194/ISPRSARCHIVES-XL-7-W2-201-2013>
 - [95] Carol Rivas. 2012. Coding and Analyzing Qualitative Data. In *Researching Society and Culture*, Clive Sale (Ed.). SAGE Publications, Thousand Oaks, CA., Chapter 21.
 - [96] Maria Riveiro, Giuliana Pallotta, and Michele Vespe. 2018. Maritime anomaly detection: A review. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery* 8, 5 (9 2018), e1266. <https://doi.org/10.1002/WIDM.1266>
 - [97] Emerald Cara Sage. 2023. Shining a light on AIS Blackouts with maritime OSINT. *Frontiers in Computer Science* 5 (8 2023), 1185760. <https://doi.org/10.3389/FCOMP.2023.1185760/BIBTEX>
 - [98] Johnny Saldaña. 2021. *The Coding Manual For Qualitative Researchers Fourth Edition*. SAGE Publications, Thousand Oaks, CA. 960 pages.
 - [99] Elizabeth B.N. Sanders. 2003. From user-centered to participatory design approaches. In *Design and the Social Sciences: Making Connections* (1st ed. ed.). CRC Press, London, Chapter 1, 1–8. <https://doi.org/10.1201/9780203301302-8/USER-CENTERED-PARTICIPATORY-DESIGN-APPROACHES-ELIZABETH-SANDERS>
 - [100] Elizabeth B.-N. Sanders and Pieter Jan Stappers. 2008. Co-creation and the new landscapes of design. *CoDesign* 4, 1 (3 2008), 5–18. <https://doi.org/10.1080/15710880701875068>
 - [101] Francisca Sasseti and Hannah Thinyane. 2021. Apprise: inclusive innovation for enhancing the agency of vulnerable populations in the context of anti-trafficking responses. *Innovation and Development* (7 2021), 1–19. <https://doi.org/10.1080/2157930X.2020.1854249>
 - [102] Seafood Watch. 2024. Seafood Social Risk Tool. <https://www.seafoodwatch.org/our-projects/seafood-social-risk-tool>
 - [103] Clive Seale. 2012. Sampling. In *Researching Society and Culture*, Clive Seale (Ed.). SAGE Publications, Thousand Oaks, CA., Chapter 9. <https://uk.sagepub.com/engb/eur/node/53906/print>
 - [104] Elizabeth R Selig, Shinnosuke Nakayama, Colette C C Wabnitz, Henrik Österblom, Jessica Spijkers, Nathan A Miller, Jan Bebbington, and Jessica L Decker Sparks. 2022. Revealing global risks of labor abuse and illegal, unreported, and unregulated fishing. *Nature Communications* 13, 1 (12 2022), 1612. <https://doi.org/10.1038/s41467-022-28916-2>
 - [105] ShipSpotting. 2024. ShipSpotting.com. <https://www.shipspotting.com/>
 - [106] Christula Stringer, D Hugh Whittaker, and Glenn Simmons. 2016. New Zealand's turbulent waters: the use of forced labour in the fishing industry. *Global Networks* 16, 1 (1 2016), 3–24. <https://doi.org/10.1111/glob.12077>
 - [107] Angelika Strohmayer, Mary Laing, and Rob Comber. 2017. Technologies and Social Justice Outcomes in Sex Work Charities: Fighting Stigma, Saving Lives. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. ACM, Denver Colorado USA, 3352–3364. <https://doi.org/10.1145/3025453.3025615>
 - [108] Hao Sun, Huanxin Zou, Kefeng Ji, Shilin Zhou, and Chunyan Lu. 2015. Combined Use of Optical Imaging Satellite Data and Electronic Intelligence Satellite Data for Large Scale Ship Group Surveillance. *The Journal of Navigation* 68, 2 (3 2015), 383–396. <https://doi.org/10.1017/S0373463314000654>
 - [109] Wilf Swartz, Andrés M Cisneros-Montemayor, Gerald G Singh, Patrick Boutet, and Yoshitaka Ota. 2021. AIS-based profiling of fishing vessels falls short as a “proof of concept” for identifying forced labor at sea. *Proceedings of the National Academy of Sciences* 118, 19 (5 2021), e2100341118. <https://doi.org/10.1073/pnas.2100341118>
 - [110] M Taconet, David Kroodsmas, and J A Fernandes. 2019. *Global Atlas of AIS-based fishing activity - Challenges and opportunities*. Technical Report. FAO, Rome. www.fao.org/3/ca7012en/ca7012en.pdf
 - [111] Renata Tesch. 2013. *Qualitative research: Analysis types and software tools*. Taylor and Francis. 1–331 pages. <https://doi.org/10.4324/9781315067339/QUALITATIVE-RESEARCH-ANALYSIS-TYPES-SOFTWARE-RENATA-TESCH>
 - [112] David Tickler, Jessica J Meeuwig, Katharine Bryant, Fiona David, John A H Forrest, Elise Gordon, Jacqueline Joudo Larsen, Beverly Oh, Daniel Pauly, Ussif R Sumaila, and Dirk Zeller. 2018. Modern slavery and the race to fish. *Nature Communications* 9, 1 (12 2018), 4643. <https://doi.org/10.1038/s41467-018-07118-9>
 - [113] UN OHCHR. 2012. *Human Rights Indicators: A Guide to Measurement and Implementation*. Technical Report. United Nations, Geneva, Switzerland. https://www.ohchr.org/sites/default/files/Documents/Publications/Human_rights_indicators_en.pdf
 - [114] John Vines, Rachel Clarke, Peter Wright, John McCarthy, and Patrick Olivier. 2013. Configuring participation: On how we involve people in design. *Conference on Human Factors in Computing Systems - Proceedings* (2013), 429–438. <https://doi.org/10.1145/2470654.2470716>
 - [115] Nick Waters, Emmanuel Freudenthal, and Logan Williams. 2020. Frontex at Fault: European Border Force Complicit in ‘Illegal’ Pushbacks. <https://www.bellingcat.com/news/2020/10/23/frontex-at-fault-european-border-force-licit-in-illegal-pushbacks/>
 - [116] Heather Welch, Tyler Clavelle, Timothy D. White, Megan A. Cimino, Jennifer Van Osdel, Timothy Hochberg, David Kroodsmas, and Elliott L. Hazen. 2022. Hot spots of unseen fishing vessels. *Science Advances* 8, 44 (11 2022), 2109. [https://doi.org/10.1126/SCIADV.ABQ2109/SUPPL\[\]FILE/SCIADV.ABQ2109\[\]SM.PDF](https://doi.org/10.1126/SCIADV.ABQ2109/SUPPL[]FILE/SCIADV.ABQ2109[]SM.PDF)
 - [117] Elizabeth White. 2023. Closing cases with open-source: Facilitating the use of user-generated open-source evidence in international criminal investigations through the creation of a standing investigative mechanism. *Leiden Journal of International Law* (2023), 1–23. <https://doi.org/10.1017/S0922156523000444>

- [118] Michael Williams and Tami Moser. 2019. The Art of Coding and Thematic Exploration in Qualitative Research. *International Management Review* 15, 1 (2019).
- [119] Nina Witjes and Philipp Olbrich. 2017. A fragile transparency: satellite imagery analysis, non-state actors, and visual representations of security. *Science and Public Policy* 44, 4 (8 2017), 524–534. <https://doi.org/10.1093/scipol/scw079>
- [120] Zhisen Yang, Zaili Yang, and Jingbo Yin. 2018. Realising advanced risk-based port state control inspection using data-driven Bayesian networks. *Transportation Research Part A: Policy and Practice* 110 (4 2018), 38–56. <https://doi.org/10.1016/j.trra.2018.01.033>

A DATA COLLECTION PROTOCOLS

The following protocol of prompts was used as a template to guide our interviews and workshops. Interviews followed the prompt schedule as reproduced below; workshops used the same prompts reformatted as a series of presentation slides.

(1) Introductions:

- (a) Please confirm your name, organisation, and job title.
- (b) How would you describe yourself and what you do?
- (c) What human rights issues do you focus on?
- (d) Which regions/jurisdictions do you focus on (if any)?
- (e) What are your areas of expertise?
- (f) Are you aware of The Alan Turing Institute and what we do? [if no, explain]
- (g) Do you feel sufficiently briefed on this project and its aims?

(2) Opening discussion:

- (a) For the purpose of clarity, we would like to check how you use some common terms. Please define:
 - Data
 - Technology
 - Human rights
 - Investigation
 - Labour exploitation
 - Migrant pushbacks
- (b) What are some examples of issues do you have focused on in the context you work in? (e.g. at a country, jurisdiction, locality, or issue-specific level)
- (c) Briefly, do practitioners in your context face challenges or regarding identifying and intervening in cases?
- (d) Briefly, what challenges do you see in your context regarding data and technology?

(3) Checking details about participants' professional experience:

- (a) Have you been involved directly in responding to any cases of rights violations at sea? E.g.,
 - Investigating
 - Intervening (e.g., trailing and boarding ships; making arrests)
 - Prosecuting
 - Supporting affected people and communities
- (b) How involved are people with lived experience of human rights abuses at sea in your work? In what ways?
- (c) Do you work with other organisations? If so, who? How?

(4) Questions about data:

- (a) What kinds of data and information sources do you or practitioners you know typically use to identify and/or intervene in rights abuses at sea?

- E.g., remote sensing data, social media data, witness testimony

(b) What do you or practitioners you know look for in the data?

(c) How do you or practitioners you know combine data from different kinds of sources?

(d) How do you or practitioners you know analyse data?

(5) Questions about technology:

(a) What kinds of tools and technologies do you or practitioners you know use to gather data?

(b) Do you or practitioners you know use formal methods and/or specific tools to analyse data?

(c) Can you think of any specific tools that you or practitioners you know use to help guide the work? How do these operate?

(d) Is there scope for more technologies to be used in this field in future? Why – or why not?

(6) Questions about workflows using data and technology:

(a) How would you or practitioners you know choose to begin a project/case?

(b) How would you or practitioners you know verify data and sources?

(c) How would you or practitioners you know analyse data once it has been gathered?

(d) How would you or practitioners you know decide if a case meets the criteria for a human rights abuse?

(e) How would you or practitioners you know account for regional differences? E.g. legal jurisdictions, economic conditions, cultural differences

(f) Are you aware of any standardised workflows or guides for conducting research and investigations at sea?

(7) Questions about ethics:

(a) What ethical issues could uses of data and tech raise in this field? For example regarding:

- Privacy
- Security
- Who has ownership and control over data or tools
- Data access
- Data misuse and related issues

(8) Final questions:

- Is there anything else you want to mention?
- Is there anything you wish to ask us?

B ADDITIONAL TABLES AND FIGURES

Type of actor	Sub-types	Examples given
Affected individuals and communities	Workers, migrants, refugees, community bodies and organisations, support networks	
Representatives and advocates	Unions, worker representatives, self-advocacy groups	International Transport Workers' Federation (ITF), Indonesian Migrant Workers Union (SBMI)
NGOs and non-profit organisations	Human rights organisations; environmentalist organisations; issue-focused campaigning organisations; investigative non-profits	Freedom Fund, Conservation International, The Labour Rights Promotion Network (LRPN), Minderoo, Forensic Architecture
INGOs	United Nations organisations, other bodies	International Labour Organisation (ILO), International Migration Organisation (IMO), International Maritime Organisation (IMO)
Academia	Universities, research institutes, independent researchers	University of Nottingham's Rights Lab
For-profit investigative organisations	News outlets, freelance journalists, private investigators	The Guardian, The Associated Press
For-profit data and technology providers	Data platform providers, database providers, data analytics providers	Lloyds of London, MarineTraffic, Ocean Mind, SkyTruth, Global Fishing Watch
Other businesses	Seafood suppliers, marine transport providers, staffing agencies	
State authorities	Government departments, statutory bodies, police forces, port authorities, militaries	Thai government, Thai military, Indonesian government

Table 3: Types of actors involved in using data to investigate human rights issues at sea

Data type	Examples	Typical sources
Analogue records	Paper-based documents	Vessels, government records
Qualitative text data	Interview transcripts, verbal testimony	Survivors, affected communities
User-generated content	Social media posts	Social media platforms, public databases
Remote sensing	AIS, VMS	Aggregators (e.g. Global Fishing Watch), governments
Satellite imagery	High-resolution satellite images, infrared spectrum imaging	Google Earth, Maxar
Surveys	Population surveys	Affected communities
Technical	System logs, internet traffic	Technical devices, public databases (e.g. Shodan)
Geolocation	GPS, co-ordinates	Vessels, mobile devices
Signals	Radio, WiFi, mobile phone signals	Vessels, mobile devices

Table 4: Data types and sources mentioned by participants

Challenge	Reasons	Example
Low quality data	Weak reporting standards; weak transparency legislation; corruption; poorly recorded data; falsification	Vessel captain deliberately falsifying records for crew working hours
Data scarcity and sparseness	Data not recorded; data no longer available	AIS data sometimes not recorded by vessels
Difficulties accessing sources	Lack of jurisdictional access; lack of resources needed to gain access; access requests refused; need to gain trust; trauma as barrier; institutional denial; corruption	Unable to access vessel records for a given country
Difficulties corroborating and verifying data	Lack of agreement between sources; lack of secondary sources	Unable to corroborate witness testimony
Difficulties analysing data	Cost; limited processing time; lack of skills	Problems paying for the compute power needed to analyse large volumes of AIS data
Duplicating data and work	Lack of coordination between organisations	Two organisations conduct the same investigation separately
Difficulties sharing data with other practitioners	Lack of trust; lack of technical standardisation; use of legal mechanisms to avoid sharing	Companies cite confidentiality to justify withholding data on activities

Table 5: Challenges and concerns relating to data

Challenge	Reasons	Example
Tools not fit for purpose or not suited for context	Poorly designed tools; inaccurate results; developers lacking knowledge of the field; tools repurposed for unsuitable contexts	An automated vessel identifier raises too many false positives; Witness reporting tool not presented in local language
Tools not accessible to those who need them most	Survivors, affected communities, and small organisations lacking resources to access tools	Fishers unable to contact support and provide testimony due to a lack of mobile phone credit
Tools not reflective of lived experiences	People with lived experience not involved in design or implementation	A statistical tool that was not developed in consultation with survivors
Bias, inaccuracies, and misinterpretation	Automation bias; confirmation bias; bias in data; users lacking technical expertise; false positives; false negatives	An over-reliance on a statistical risk-evaluation tool without ground-truthing the results in the experienced of affected communities
Over-saturated market	Lots of developers and vendors offering competing tools	Multiple vendors selling extremely similar 'AI' vessel detection tools
Tech solutionism	An assumption that technology will be able to solve any problem	Misguided attempts to apply blockchain technology to supply chains assuming it will identify exploitation

Table 6: Challenges and concerns relating to tools and technologies